

Lab Report: CLIP-LoRA on the Walnut Dataset

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Repository: <https://github.com/myavarih/CLIP-LoRA>

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1. How Does Adaptation Work

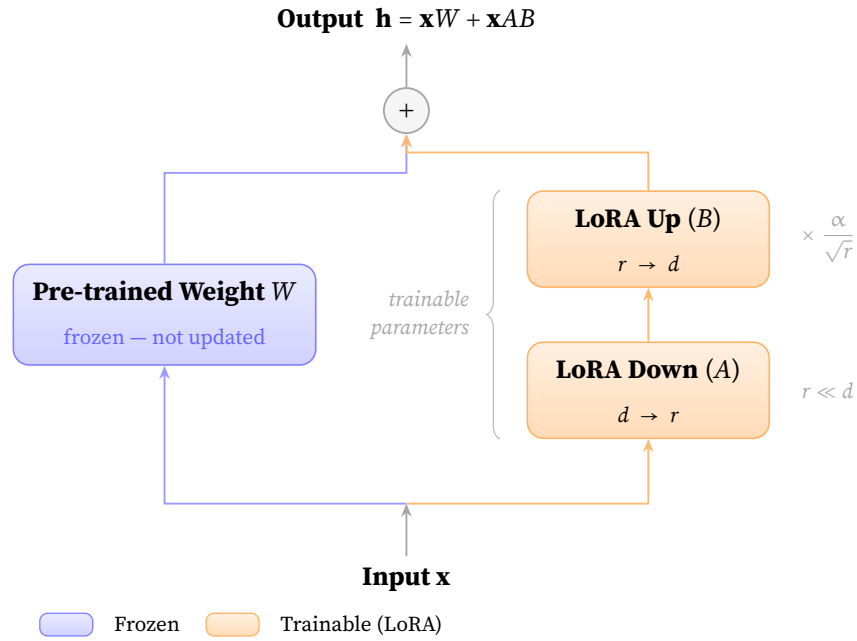


FIGURE 1. This figure shows an overview of how learnable parameters are injected in a layer.

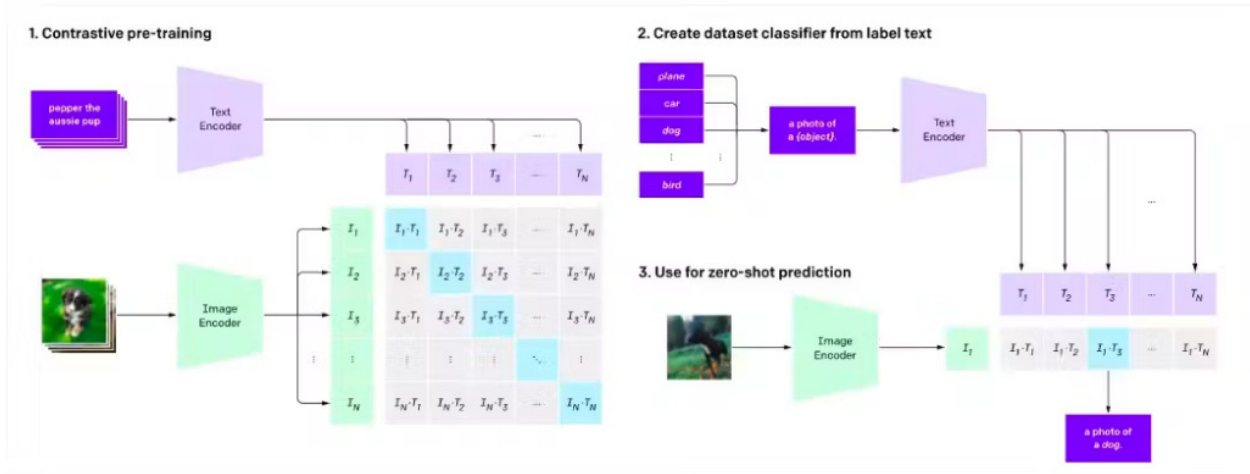


FIGURE 2. How CLIP classifies.

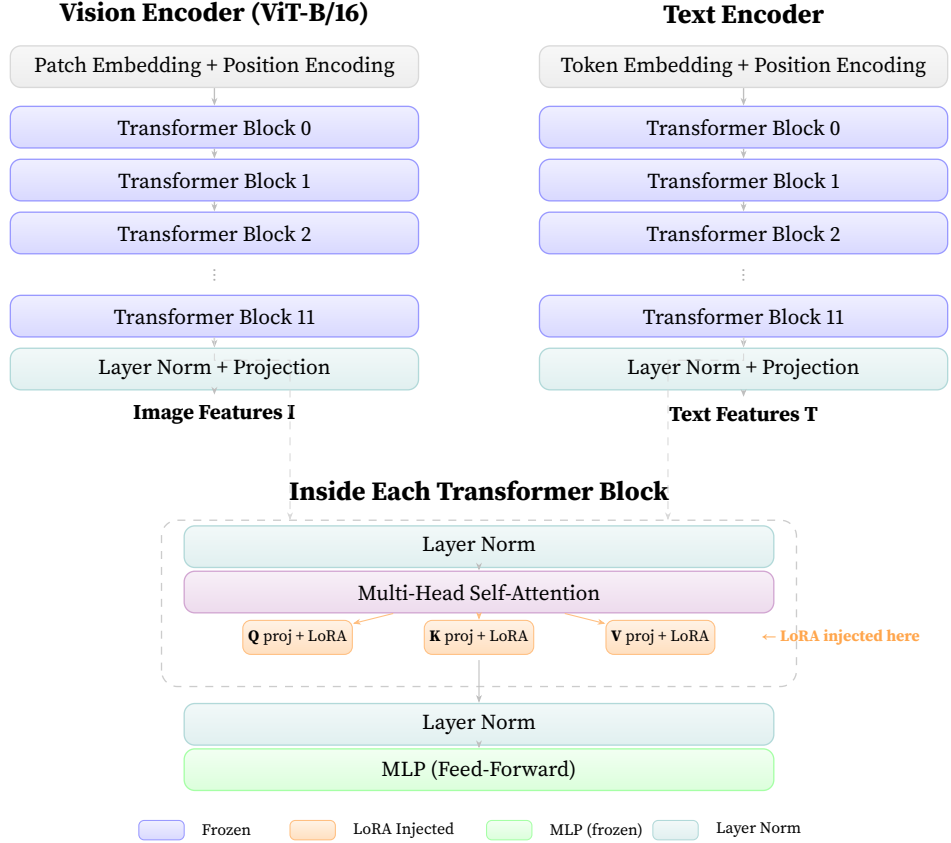


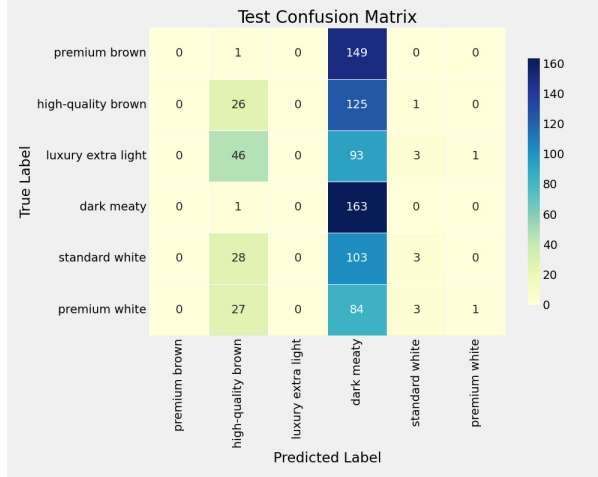
FIGURE 3. Internal architecture of CLIP’s vision and text encoders (ViT-B/16). Both encoders have 12 transformer blocks. LoRA parameters are injected into the Q, K, and V projections inside the multi-head self-attention of each block.

2. Visualizations (Added)

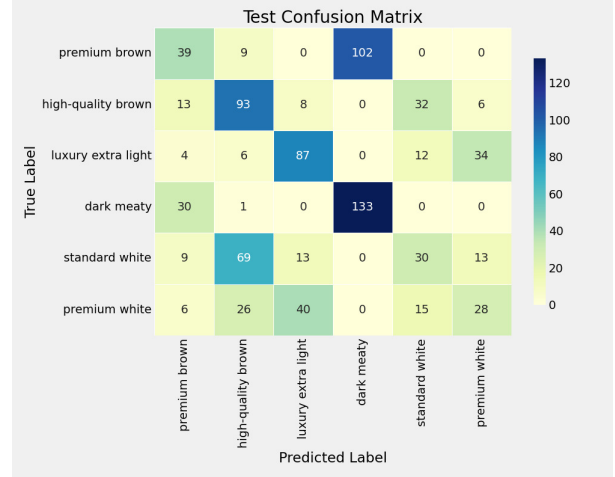
To better understand model behavior beyond overall accuracy metrics, confusion matrices, sample predictions, t-SNE, PCA, and EigenCAM attention visualizations were added across both train and test sets.

2.1. Confusion Matrices

Confusion matrices display class-level misclassifications. Diagonal values show correct predictions, while off-diagonal values show where classes get confused.



A. Zero-shot (before LoRA)

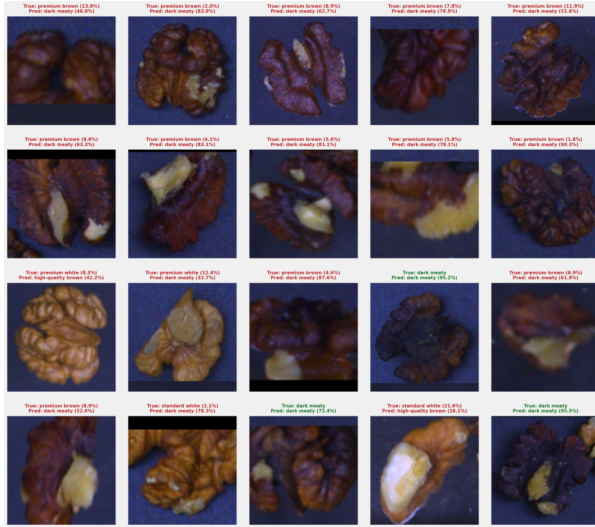


B. After LoRA fine-tuning (4-shot)

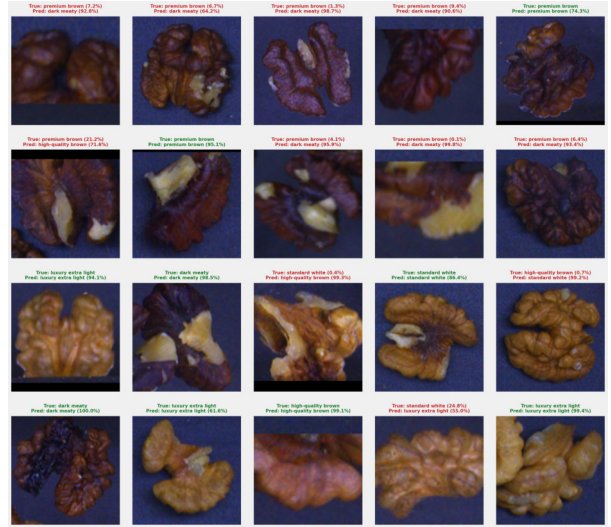
FIGURE 4. Confusion matrices on the test set before and after LoRA adaptation.

2.2. Sample Predictions

Sample predictions show test images with predicted class labels and confidence scores.



A. Zero-shot predictions



B. After LoRA fine-tuning (4-shot)

FIGURE 5. Sample test predictions with confidence scores.

2.3. t-SNE and PCA

PCA and t-SNE are used to project high-dimensional embeddings into 2D:

- **PCA** preserves global variance across features.
- **t-SNE** preserves local neighbor distances to reveal distinct class clusters.

Separated clusters show clear features; overlapping clusters indicate class ambiguity.

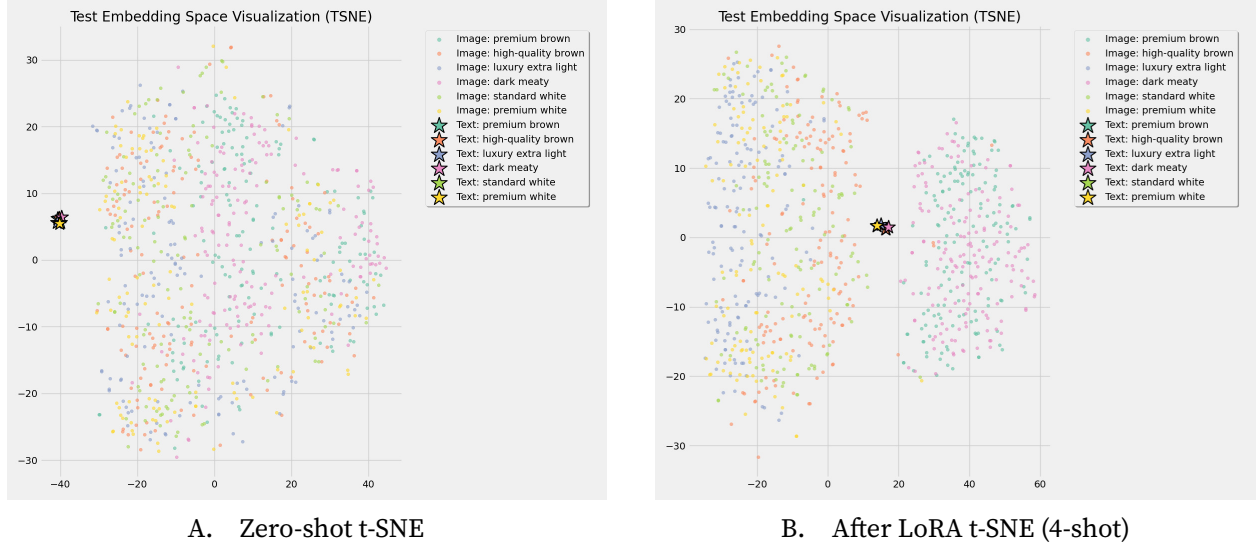


FIGURE 6. t-SNE visualizations of test embeddings before and after LoRA.

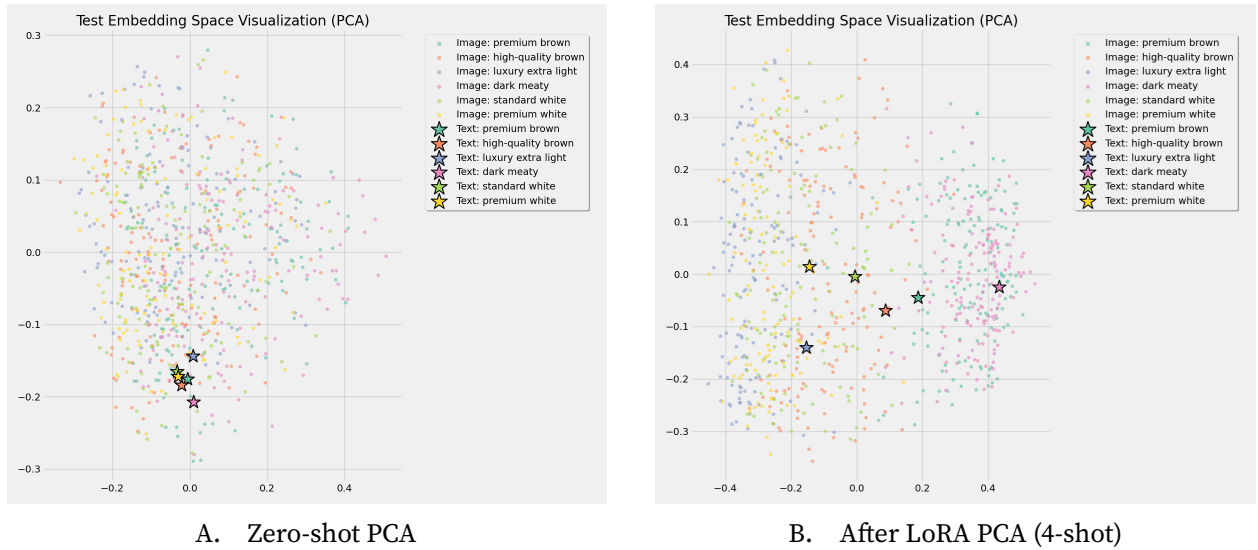


FIGURE 7. PCA visualizations of test embeddings before and after LoRA.

2.4. Attention Visualizations via EigenCAM

Standard Grad-CAM struggles here for three main reasons:

- **No classification head:** CLIP uses cosine similarity between image and text features rather than softmax logits.
- **Frozen weights:** Backpropagating through frozen layers yields weak attributions since only LoRA layers update.

- **Mixed precision (FP16):** Half-precision gradients often generate noisy heatmaps.

EigenCAM avoids this by being gradient-free. It computes the **first principal component** of activations at the target layer—compressing 768 feature channels into a 2D heatmap of the primary visual patterns. This reveals whether the model focuses on the walnut itself or background clutter.

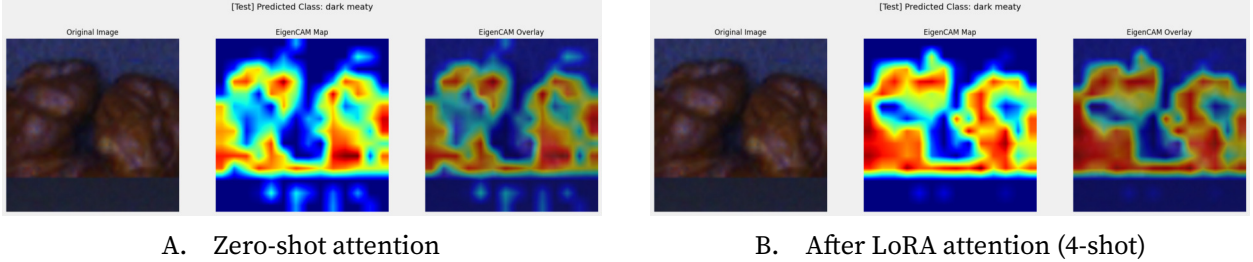


FIGURE 8. EigenCAM attention maps on a test image before and after LoRA adaptation.

3. Hyperparameter Explanation

TABLE 1. Fixed baseline hyperparameters across experiments.

Hyperparameter	Default Value	Purpose
Backbone	ViT-B/16	Pre-trained feature extractor
Learning rate	2×10^{-4}	Gradient step size
Batch size	32	Samples per iteration
Iterations	500 (per shot)	Total optimization steps
LoRA rank (r)	2	Low-rank inner dimension
LoRA scaling (α)	1.0	Adapter output multiplier
Dropout rate	0.25	Regularization rate
Target parameters	q, k, v	Attention projection targets
Target encoder	Both	Encoder modality targeting
Layer position	All layers	Layer depth targeting

4. Zero-Shot Baseline Analysis

Initial zero-shot testing yielded an accuracy near 19%. Modifying the prompt templates and translating class names into descriptive English improved accuracy to 22.26%. For example:

a photo of a brown momtaz walnut → a photo of a premium brown
walnut, a type of walnut.

Despite this improvement, zero-shot accuracy remains very low and close to random guessing. CLIP’s pre-training did not include fine-grained walnut cultivars, so its default embeddings cannot distinguish between these varieties without fine-tuning.

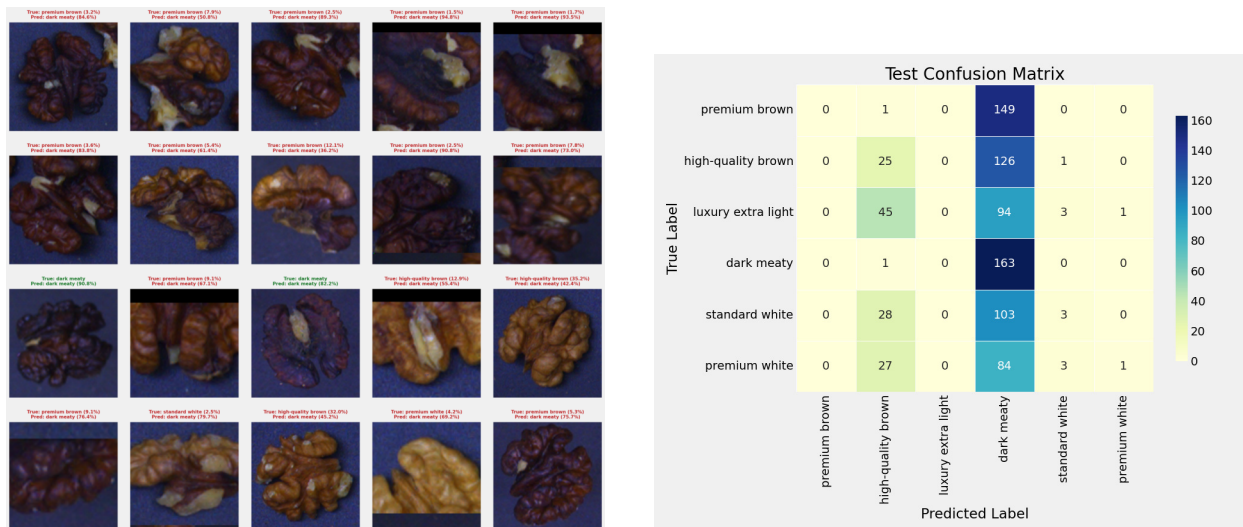


FIGURE 9. Zero-shot test predictions and confusion matrix.

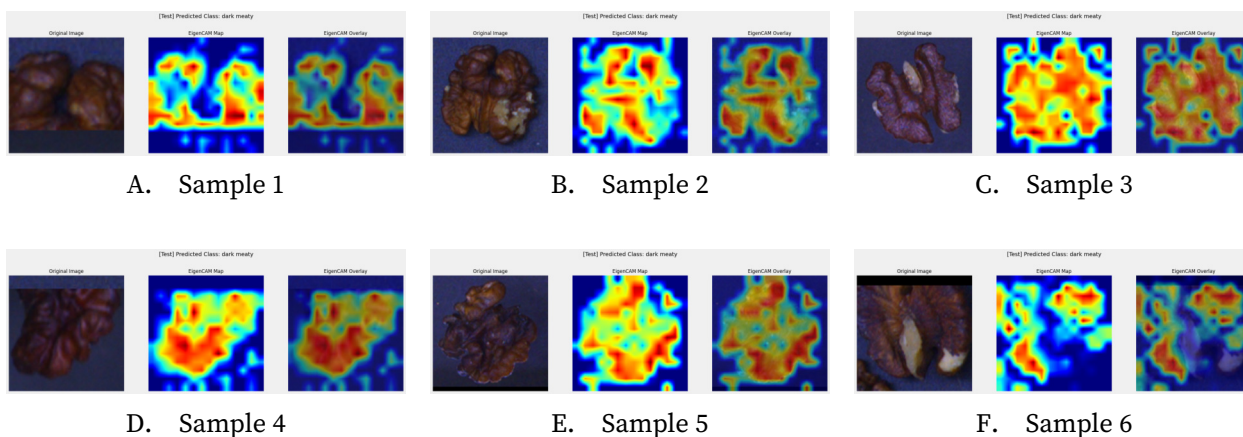


FIGURE 10. Zero-shot EigenCAM attention maps across 6 test samples.

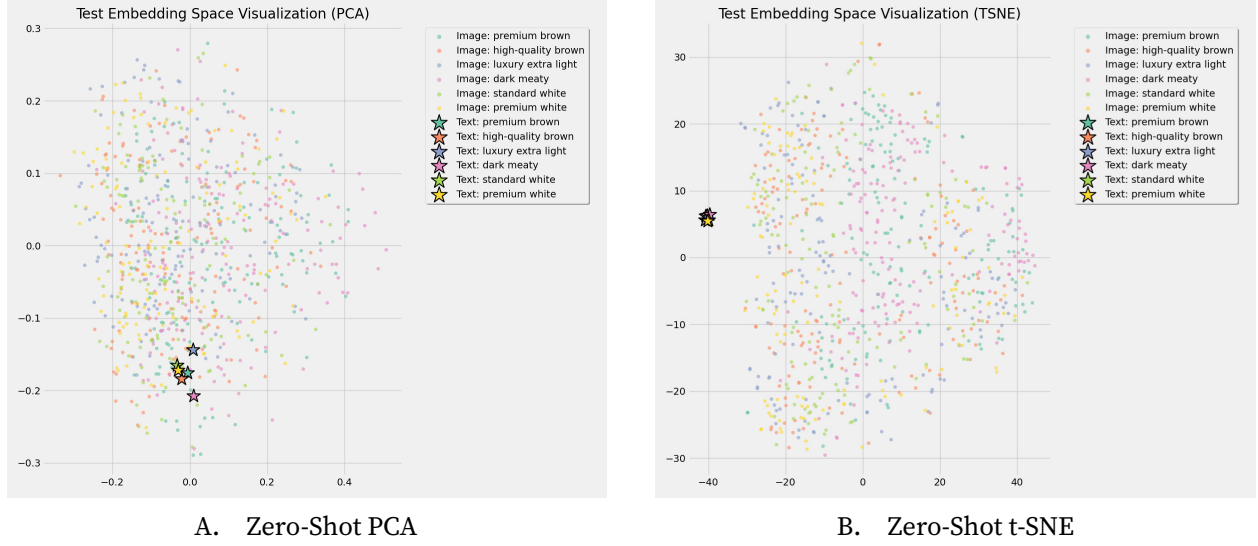


FIGURE 11. PCA and t-SNE projections of zero-shot test embeddings.

5. The Numbers: Complete Ablation Results

Table 2 summarizes test accuracy across all configurations and shot counts. Figure 12 compares overall accuracy progression. Testing all combinations was not feasible, so a representative subset was selected.

TABLE 2. Comprehensive test accuracy (%) across all experiments. **Green** = best per column, **Red** = worst. Missing values (-) = not evaluated.

Configuration	Seed	Number of Shots						Train Acc (4s)	Time (4s)
		1	2	4	8	16	32		
Target Layer Ablations ($r = 2, \alpha = 1.0, d = 0.25$)									
baseline (q, k, v)	1	39.74	43.82	46.39	49.65	56.18	62.82	100.0	30m
baseline (q, k, v)	2	38.93	43.82	47.67	50.70	59.32	–	100.0	29m
baseline (q, k, v) ablation run	2	38.46	44.06	48.02	50.35	59.56	–	100.0	29m
p_qv (q, v)	2	36.60	41.26	44.87	52.80	58.51	–	100.0	–
p_v (v only)	2	40.68	40.21	46.97	50.00	58.16	–	100.0	–
r_1 ($q, k, v, r = 1$)	2	38.23	43.12	47.55	49.42	–	–	100.0	–
Alpha 1.5 Sweep ($r = 2, \alpha = 1.5, d = 0.25$, Seed 3)									
baseline_α1.5 (q, k, v)	3	39.39	42.31	49.65	51.63	58.28	–	100.0	27m
p_qv_α1.5 (q, v)	3	39.63	42.66	48.14	50.70	58.62	–	100.0	29m
p_v_α1.5 (v only)	3	38.69	41.49	47.32	50.93	58.51	–	100.0	29m
r_1_α1.5 ($r = 1$)	3	38.11	42.77	48.02	50.47	–	–	100.0	32m
High Alpha & Dropout Experiments									
α=2.0, d=0.4 (q, k, v)	1	39.98	41.61	44.29	53.61	–	–	100.0	15m
α=1.5, d=0.4 (q, k, v)	1	40.56	43.01	44.17	52.68	–	–	95.83	15m
4-Shot Only Ablations ($\alpha = 1.0, d = 0.25$, Seed 2)									
a_0.75 ($\alpha = 0.75$)	2	–	–	47.32	–	–	–	100.0	–
a_1.5 ($\alpha = 1.5$)	2	–	–	50.00	–	–	–	100.0	–
drop_0.4 (Dropout 0.4)	2	–	–	48.37	–	–	–	100.0	–
e_text (Text Encoder)	2	–	–	40.56	–	–	–	100.0	–
e_vision (Vision Encoder)	2	–	–	45.34	–	–	–	100.0	–
r_3 (Rank 3)	2	–	–	48.60	–	–	–	100.0	–

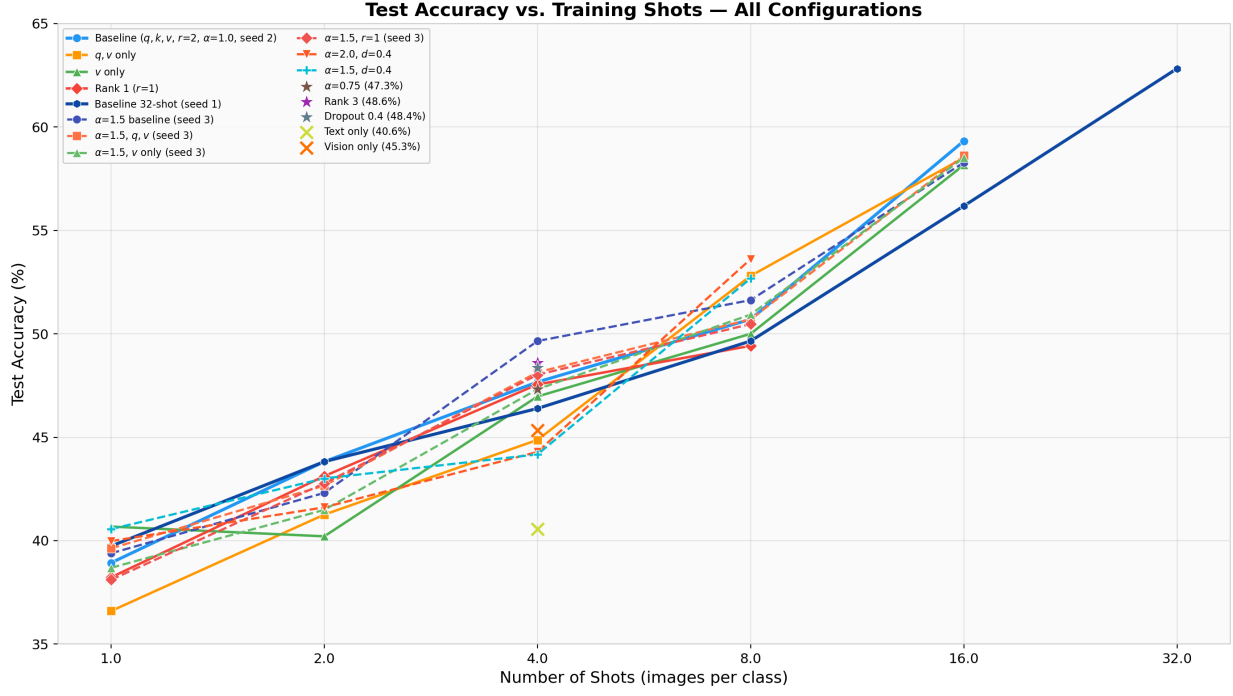


FIGURE 12. Test accuracy vs. shots across all configurations.

5.1. Cross-Seed Variance

Evaluating the baseline across random seeds shows low standard deviation ($\approx \pm 1\text{--}2\%$).

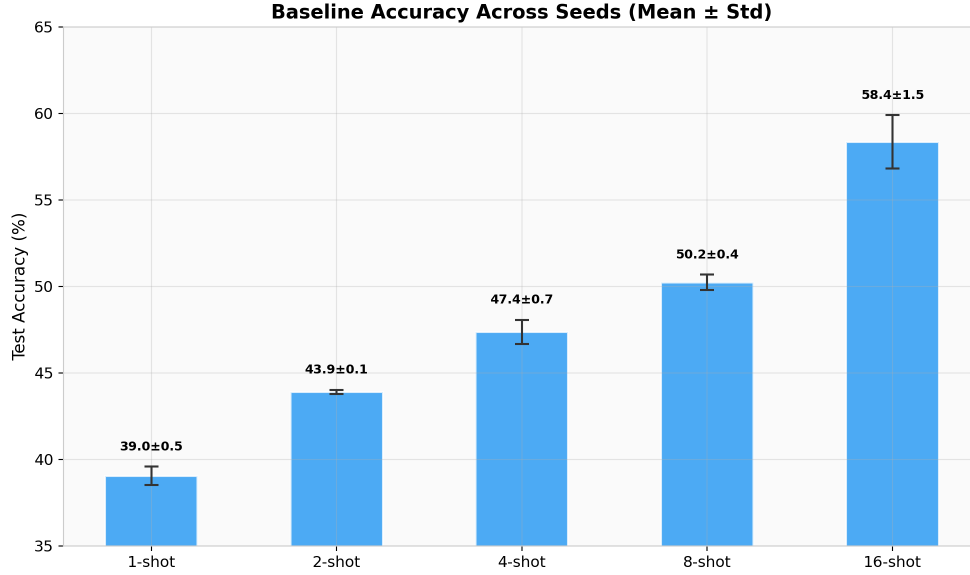


FIGURE 13. Baseline accuracy across seeds (mean $\pm$ std).

6. Training Dynamics

6.1. Training Curves (4-Shot)

Figure 14 tracks test accuracy over 2,000 iterations for all 4-shot models.

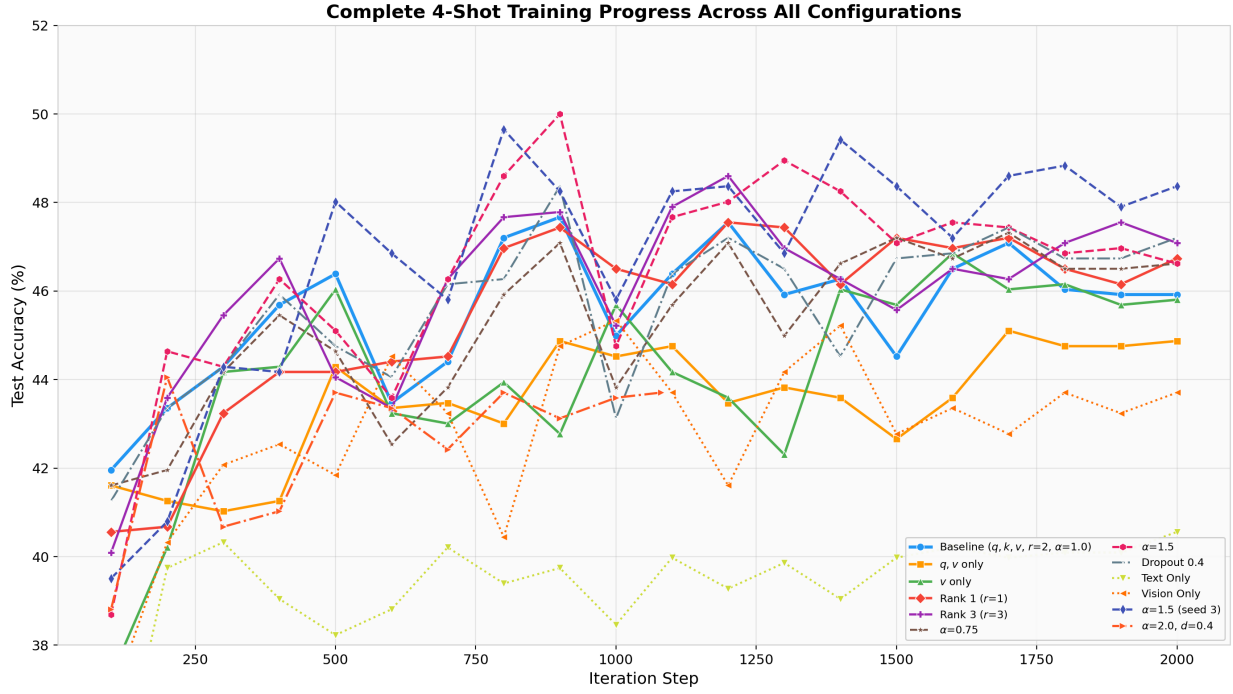


FIGURE 14. Complete 4-shot training progress across all 12 configurations evaluated over 2,000 iterations.

6.2. Convergence Speed & Training Time

Training time increases almost linearly with shot count (Figure 15).

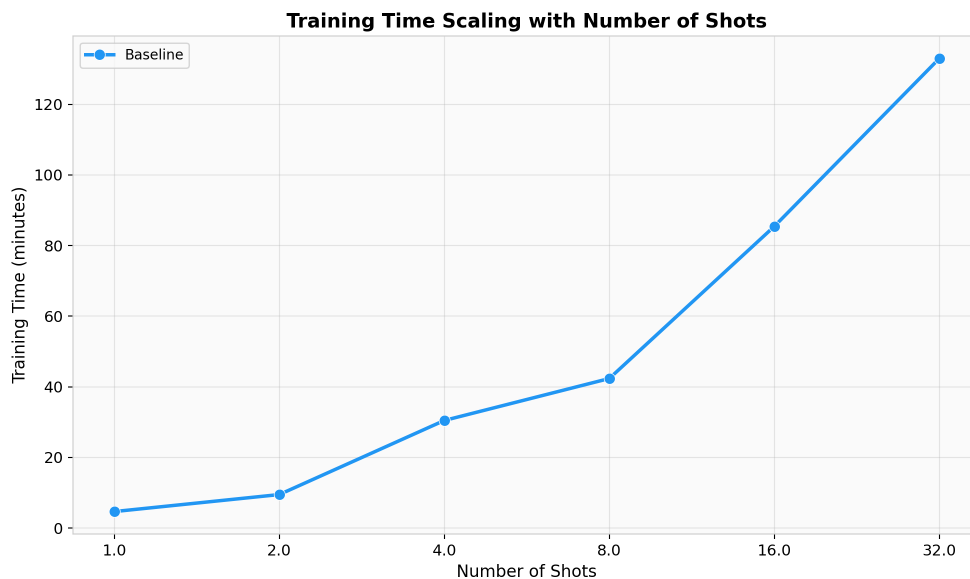


FIGURE 15. Training time scaling across shot counts.

6.3. Overfitting Analysis

Extreme overfitting is visible in all shots (because it is very easy for the huge model to memorize a few train images), but more shots reduce the gap as expected (Figure 16).

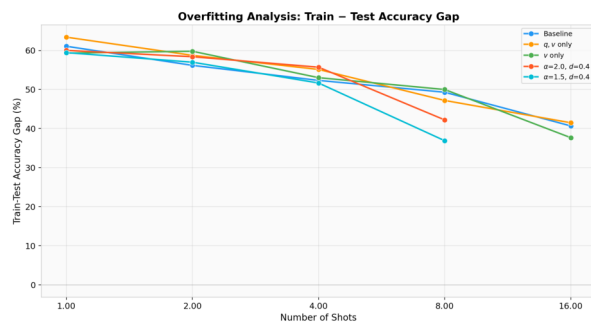
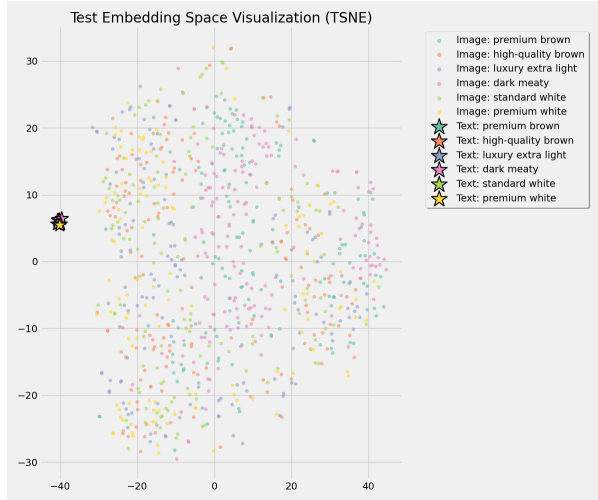


FIGURE 16. Train-test accuracy gap across shot counts.

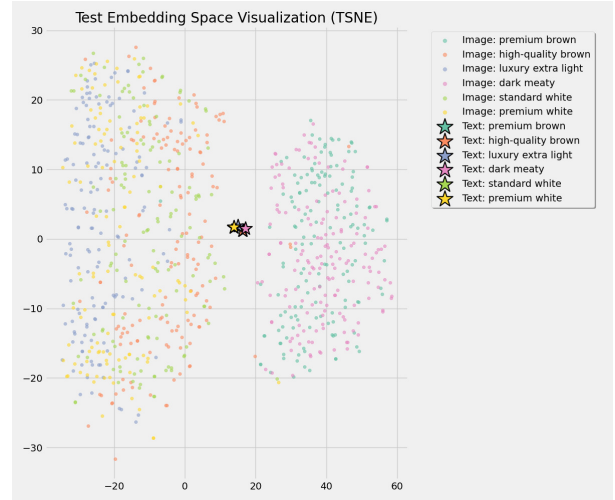
7. Deep Dive: Before vs. After (4-Shot)

7.1. Clustering

Hints of clustering are visible (Figure 17 & Figure 18).

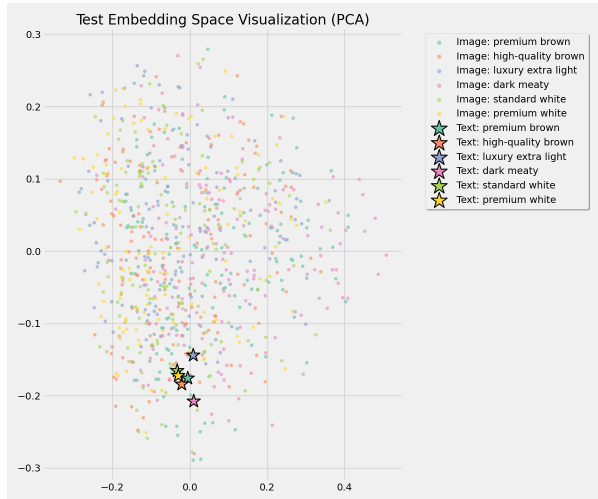


A. Zero-Shot

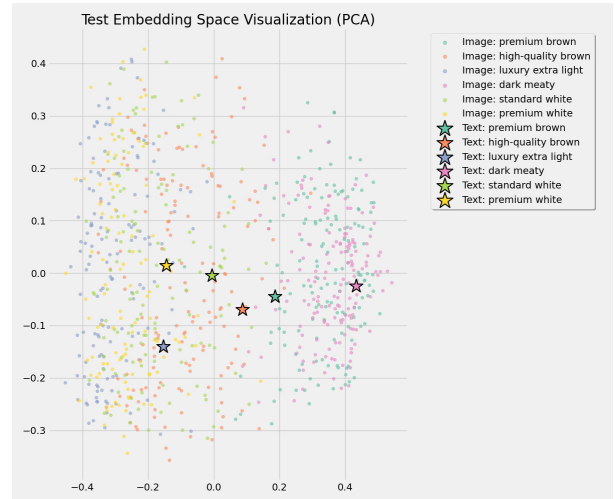


B. 4-Shot Baseline

FIGURE 17. t-SNE comparison of test embeddings before and after 4-shot LoRA.



A. Zero-Shot

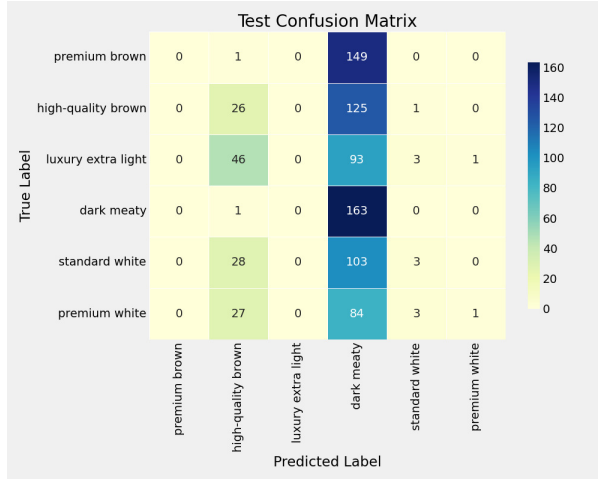


B. 4-Shot Baseline

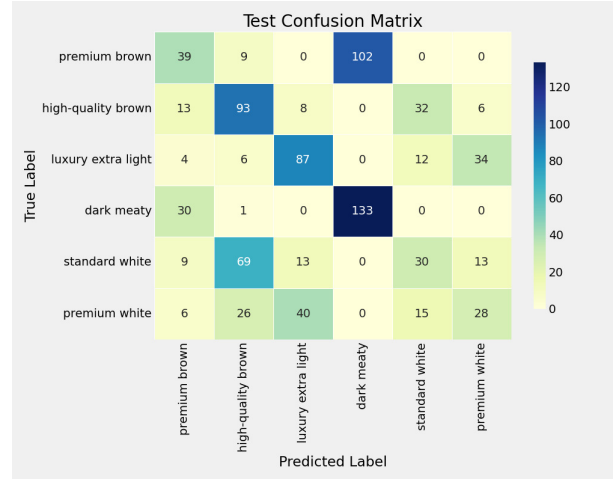
FIGURE 18. PCA comparison of test embeddings before and after 4-shot LoRA.

7.2. Predictions and Confusion

Visualizations illustrate model learning dynamics while highlighting where and why failure cases occur (Figure 19 & Figure 20).

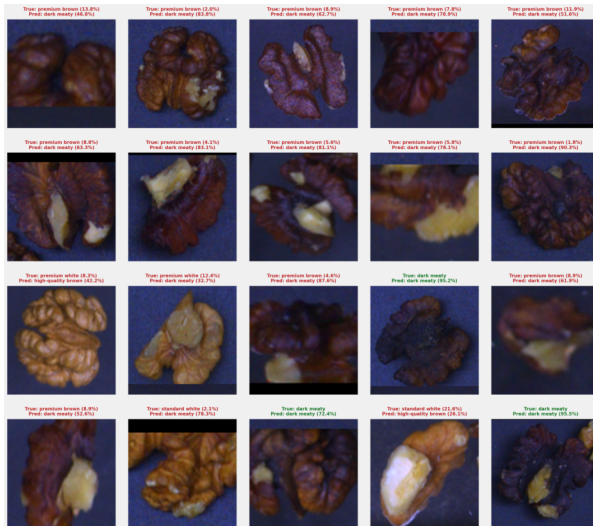


A. Zero-Shot Confusion

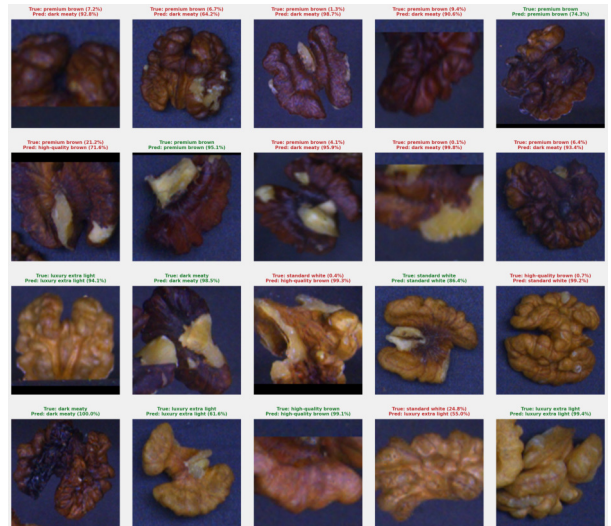


B. 4-Shot Confusion

FIGURE 19. Confusion matrix before and after 4-shot LoRA adaptation.



A. Zero-Shot Predictions



B. 4-Shot Predictions

FIGURE 20. Sample predictions before and after 4-shot LoRA adaptation.

7.3. Attention Map Shift

Attention focus shows slight spatial refinement (Figure 21).

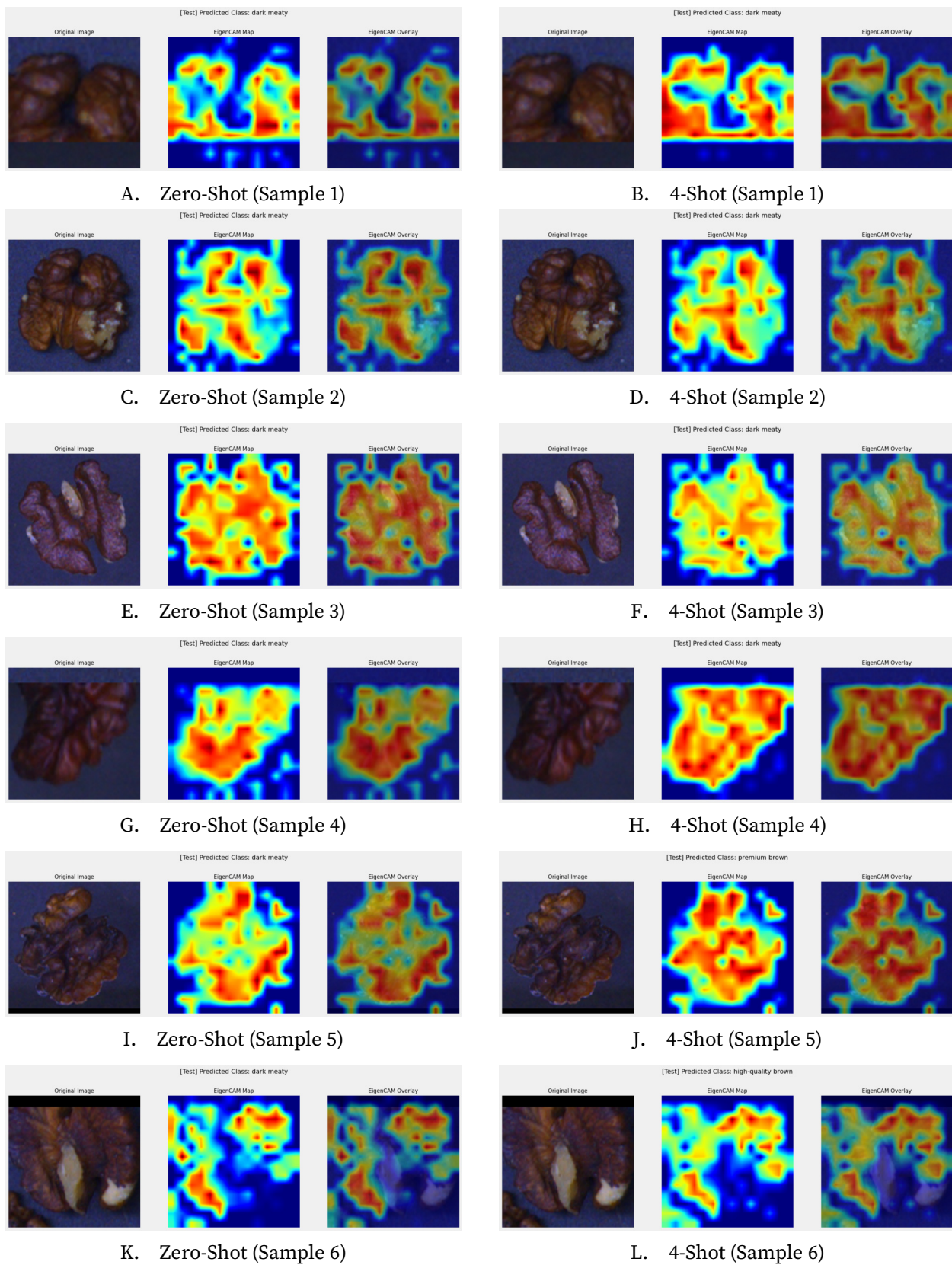


FIGURE 21. EigenCAM attention comparison showing focus shift to visual features across 6 samples (Zero-Shot vs. 4-Shot Baseline).

8. Advanced Ablations

8.1. Alpha Scaling Sweep

Evaluating different scaling factors ($\alpha \in \{0.75, 1.0, 1.5, 2.0\}$) indicates that $\alpha = 1.5$ achieves peak performance at 4-shot (Figure 22 & Figure 23). However, across other shot counts, no single configuration works optimally in all cases (even within the same dataset). See Figure 24 for target layer variations under $\alpha = 1.5$.

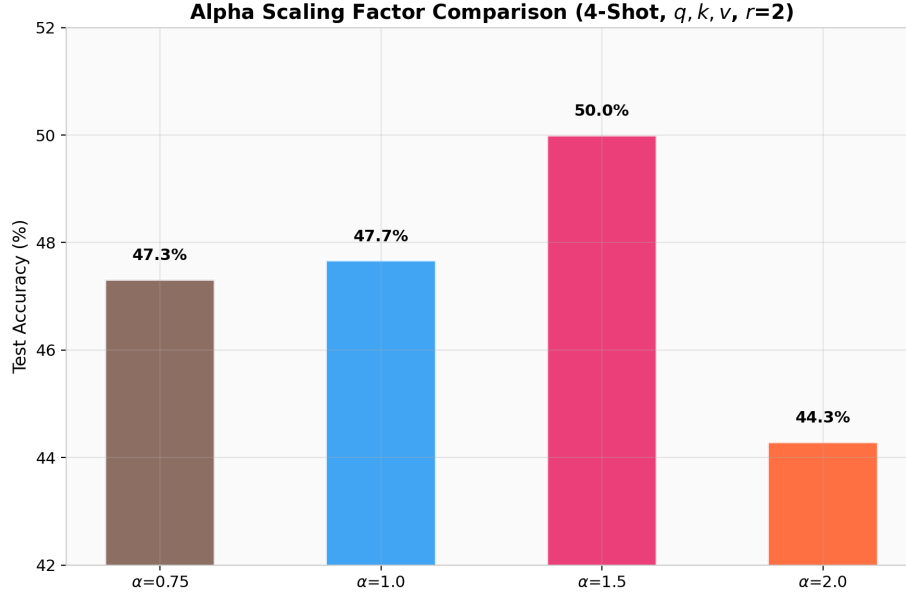


FIGURE 22. Alpha scaling comparison at 4-shot.

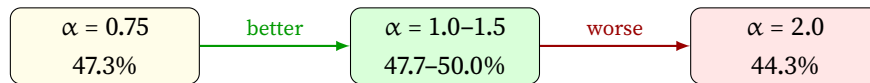


FIGURE 23. Alpha scaling performance summary.

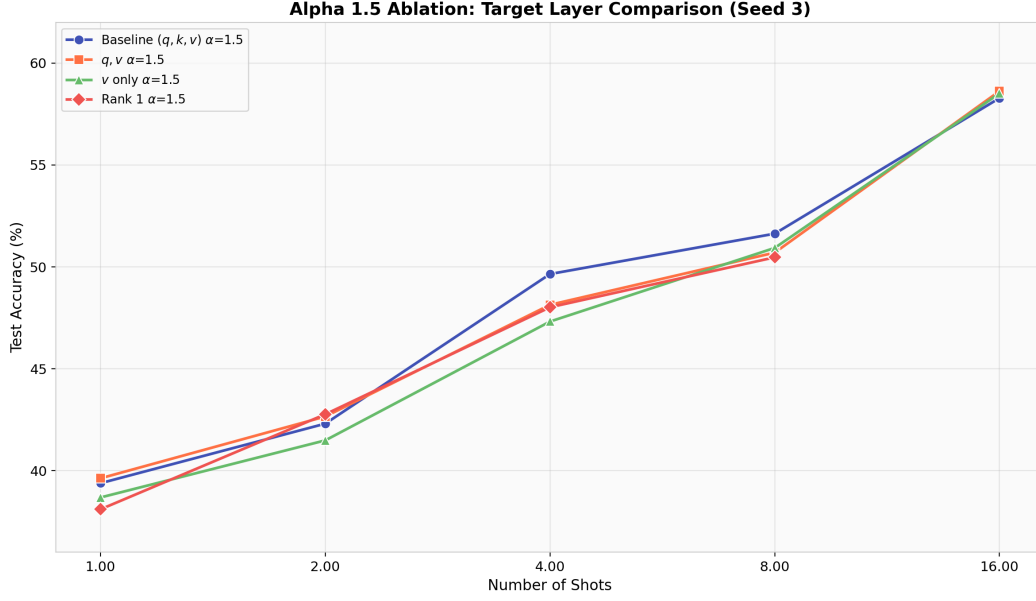


FIGURE 24. $\alpha = 1.5$ ablation grid across target layer configurations (Seed 3).

8.1.1. Alpha 2.0 Analysis

At $\alpha = 2.0$, 4-shot performance drops to 44.3%, but recovers at 8-shot to achieve 53.6%.

As mentioned, finding a single configuration that consistently outperforms the baseline across all shot counts turned out to be a dead end. However, increasing α grants the model greater adaptation power ("wilder" parameter updates), though it also makes performance less predictable. Tuning seeds alongside these hyperparameters may yield slightly superior results in specific isolated cases.

8.2. Rank Analysis

Most tests were conducted with $r = 2$. Although $r = 1$ is less likely to yield improvements, $r = 3$ may be worth investigating further as it increases adaptation power (Figure 25–27).

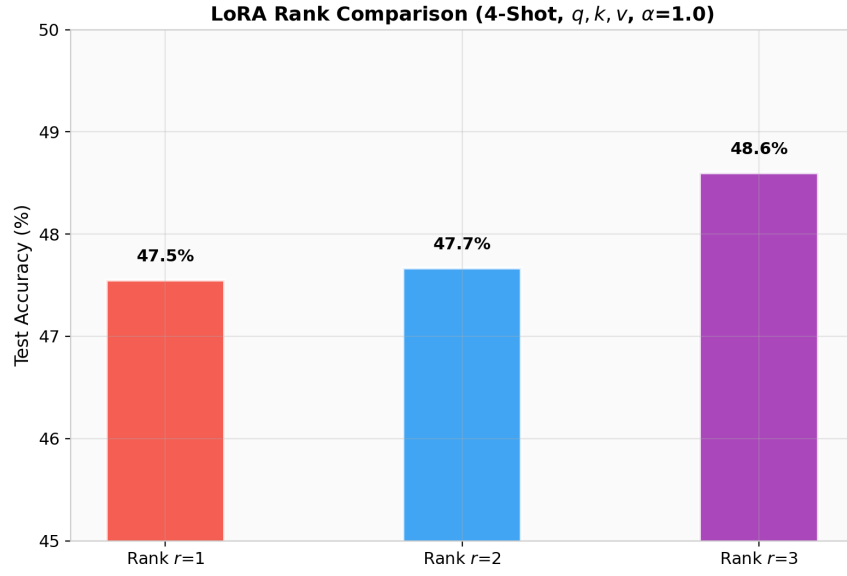


FIGURE 25. LoRA rank comparison at 4-shot.

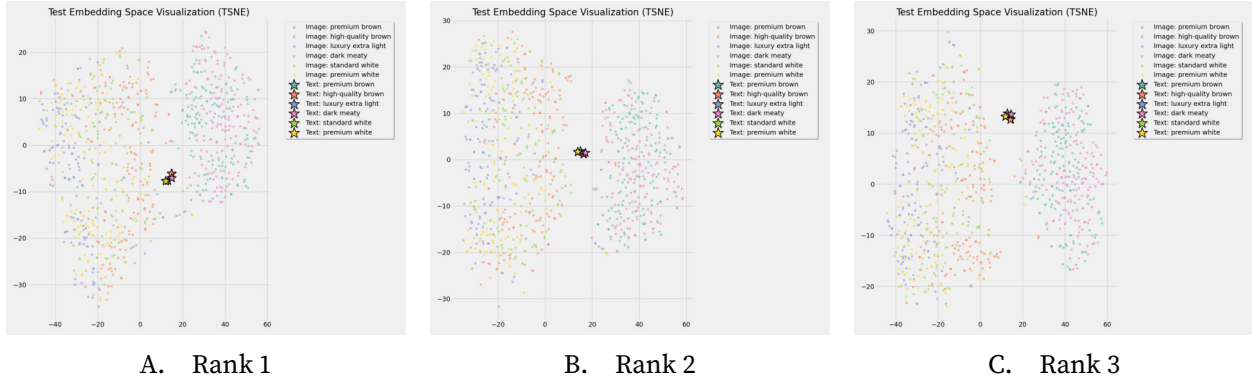


FIGURE 26. t-SNE comparison across LoRA ranks.

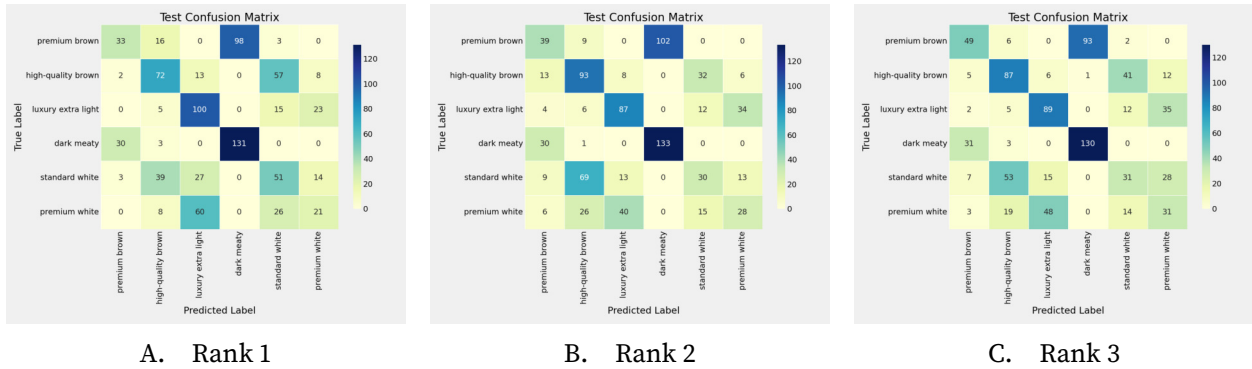


FIGURE 27. Confusion matrices across LoRA ranks.

8.3. Modality Targeting: Text vs. Vision

Adaptation power is maximized when LoRA parameters are injected into both encoders (Figure 28–30).

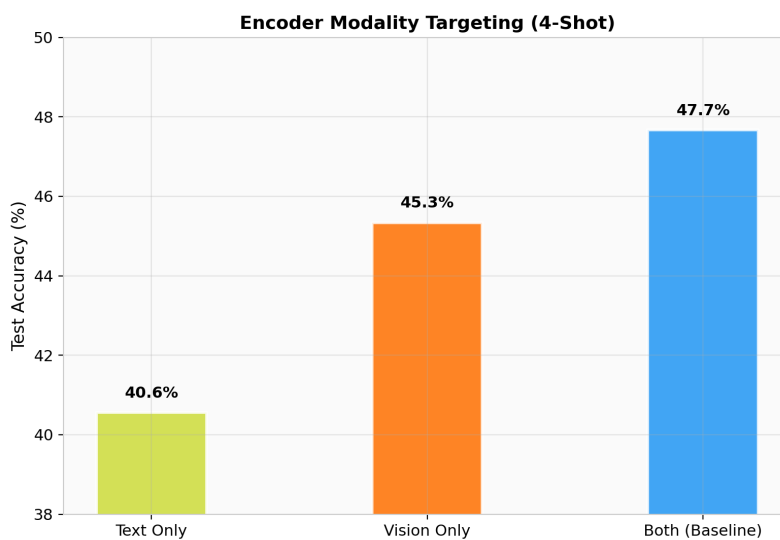


FIGURE 28. Encoder modality targeting comparison at 4-shot.

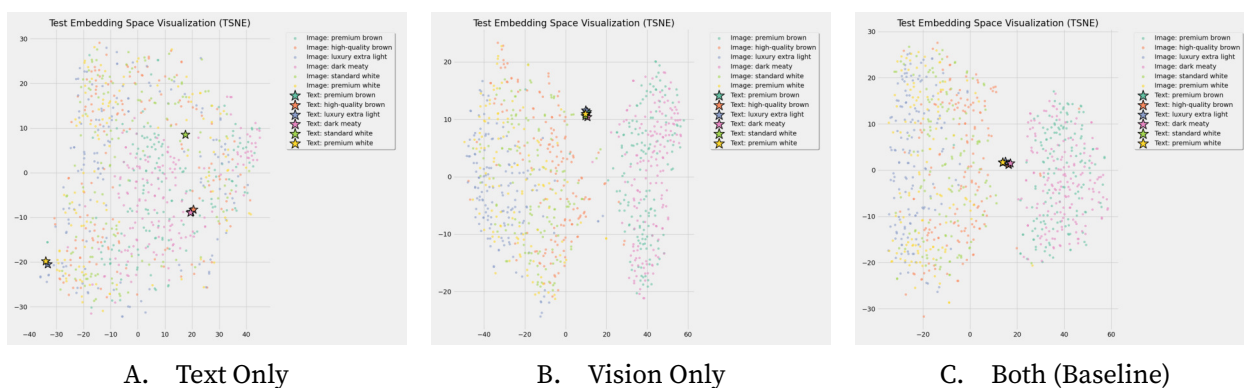


FIGURE 29. Feature space comparison by encoder targeting.

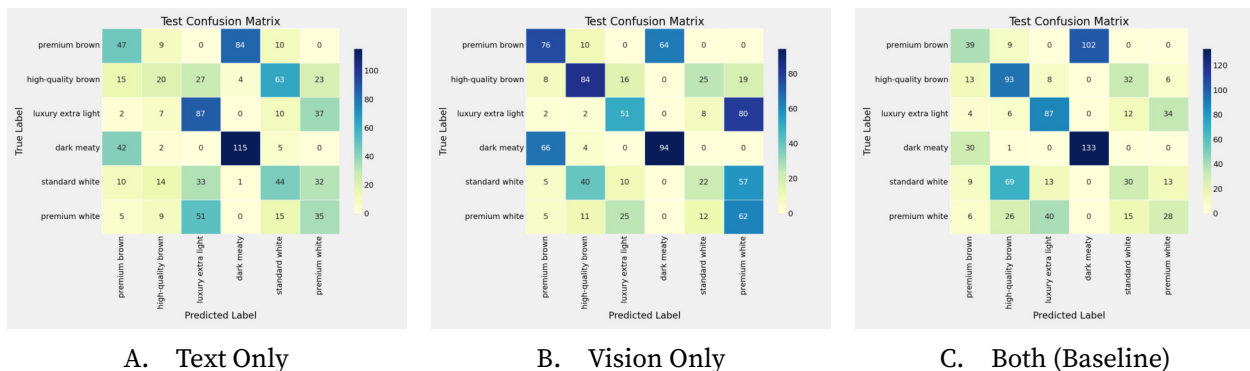


FIGURE 30. Confusion matrices by encoder targeting.

9. Scaling: From 1-Shot to 32-Shot

Baseline accuracy scales consistently from 39.7% at 1-shot to 62.82% at 32-shot (Figure 31).

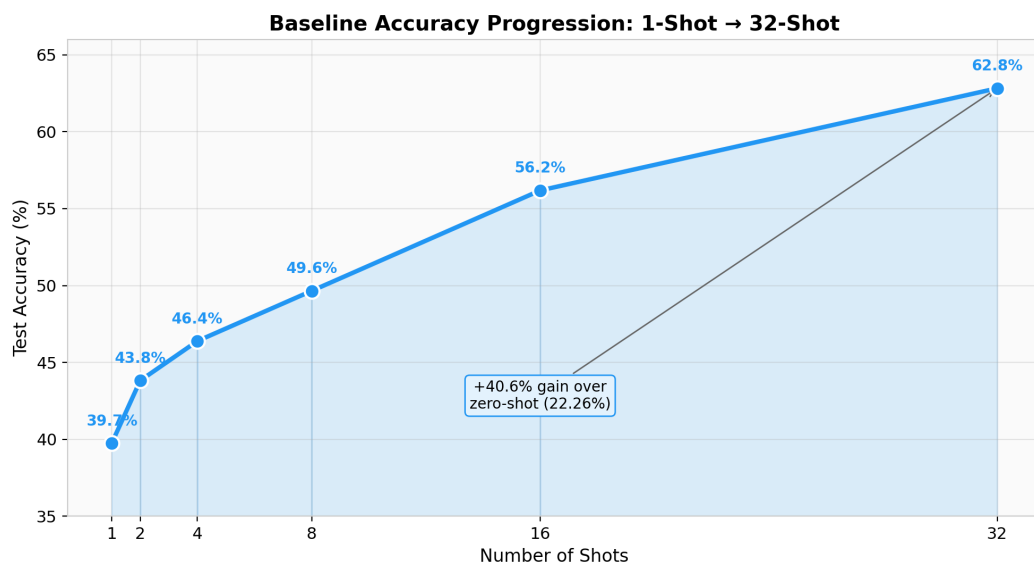
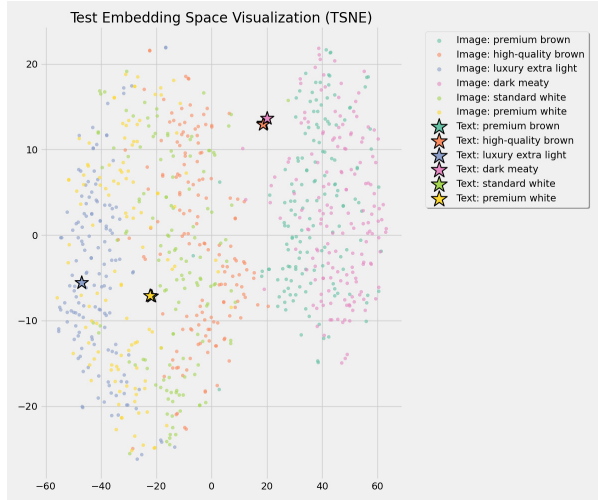


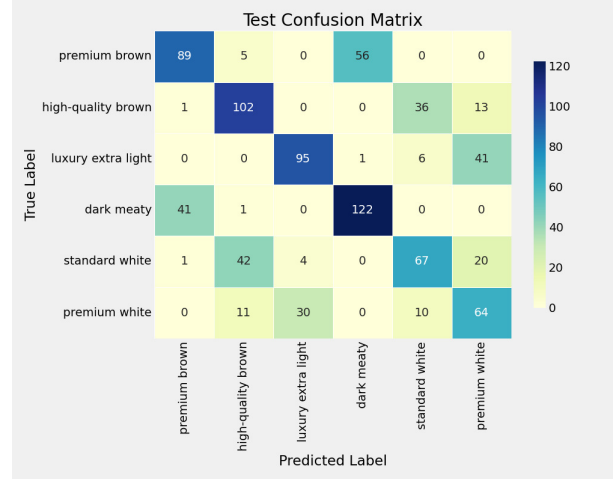
FIGURE 31. Baseline accuracy scaling from 1-shot to 32-shot.

9.1. 32-Shot Evaluation

At 32 shots per class, embeddings form tight clusters and confusion matrix errors decrease substantially (Figure 32 & Figure 33).

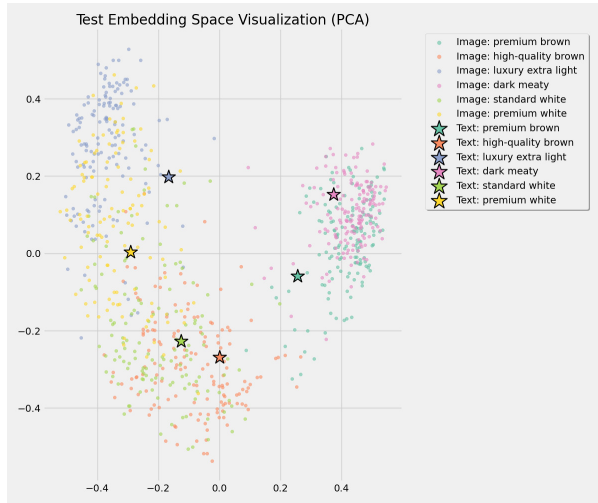


A. 32-Shot t-SNE

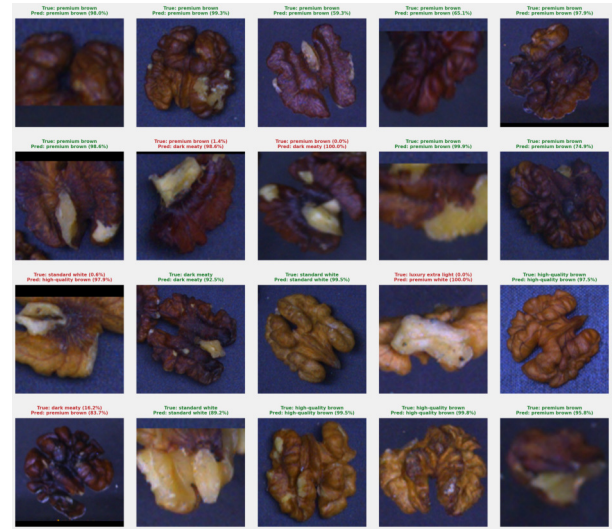


B. 32-Shot Confusion Matrix

FIGURE 32. 32-shot baseline t-SNE and confusion matrix.



A. 32-Shot PCA



B. 32-Shot Predictions

FIGURE 33. PCA and sample predictions at 32-shot.

9.2. Attention Evolution across Shots

EigenCAM maps show progressive spatial focus refinement from zero-shot all the way to 32-shot adaptation (Figure 34).

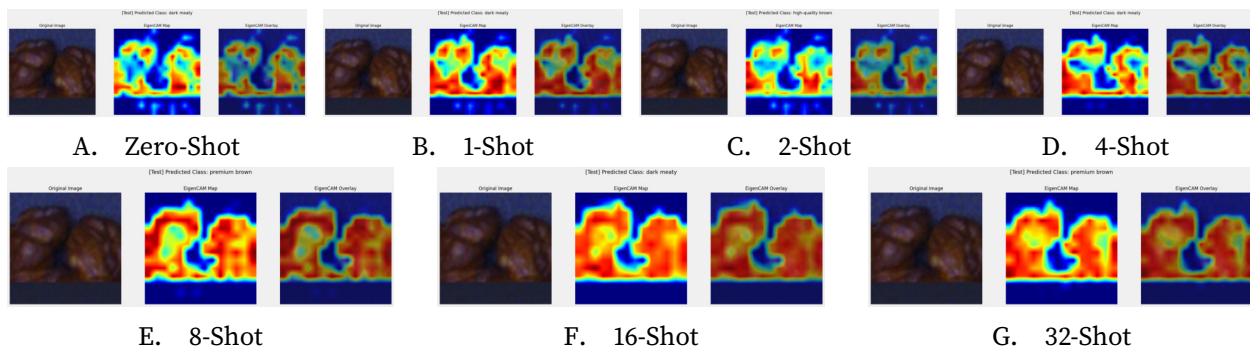


FIGURE 34. Attention map progression across shot counts (Zero-Shot to 32-Shot).

10. Catastrophic Forgetting & Generalization

Evaluating the adapted models on ImageNet validation demonstrates minimal catastrophic forgetting:

- **4-Shot Model:** Achieves **66.66%** ImageNet validation accuracy (compared to 66.71% zero-shot CLIP, a negligible 0.05% drop).
- **32-Shot Model:** Achieves **64.78%** ImageNet validation accuracy (evaluation time: 176.53s, a 1.93% drop).

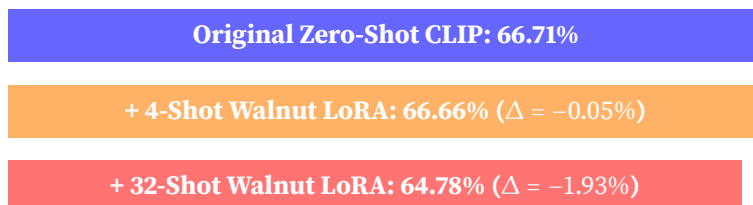


FIGURE 35. ImageNet validation accuracy across zero-shot, 4-shot, and 32-shot models.

The forgetting is not significant. This is promising and suggests multiple fine-grained tasks can be learned sequentially without catastrophic forgetting (as discussed in the C-CLIP paper).